

THE RESEARCH STAGES IN CONDUCTING A LITERATURE REVIEW



Stage Characteristics	Research stage				
	<i>Problem formation</i>	<i>Data collection</i>	<i>Data evaluation</i>	<i>Analysis and interpretation</i>	<i>Public presentation</i>
Research questions asked	What evidence should be included in the review?	What procedures should be used to find relevant evidence?	What retrieved evidence should be included in the review?	What procedures should be used to make inferences about the literature as a whole?	What information should be included in the review report?
Primary function in review	Constructing definitions that distinguish relevant from irrelevant studies.	Determining which sources of potentially relevant sources to examine.	Applying criteria to separate "valid" from "invalid" studies.	Synthesizing valid retrieved studies.	Applying editorial criteria to separate important from unimportant information.
Procedural differences that create variation in review conclusion	1. Differences in included operational definitions. 2. Differences in operational detail.	Differences in the research contained in sources of information.	1. Differences in quality criteria. 2. Differences in the influence of non-quality criteria.	Differences in the rules of inference.	Differences in guidelines for editorial judgment.
Sources of potential invalidity in review conclusions	1. Narrow concepts might make review conclusions less definitive and robust. 2. Superficial operational detail might obscure interacting variables.	1. Accessed studies might be qualitatively different from the target population of studies. 2. People sampled in accessible studies might be different from target population of people.	1. Nonequality factors might cause improper weighting of study formation. 2. Omissions in study reports might make conclusions unreliable.	1. Rules for distinguishing patterns from noise might be inappropriate. 2. Review-based evidence might be used to infer causality.	1. Omission of review procedures might make conclusions irreproducible. 2. Omission of review findings and study procedures might make conclusions obsolete.
5. Rhetoric	L. Was written with a coherent, clear structure that supported the review	Poorly conceptualized, haphazard	Some coherent structure	Some coherent structure	Well developed, coherent

From "Scientific Guidelines for Conducting Integrative Research Reviews," *Review of Education Research*, 1984, 52, pg. 293. Copyright 1984 by Sage Publications. Reprinted with permission of Sage Publications.

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Source: Randolph, Justus (2009). A Guide to Writing the Dissertation Literature Review. Practical Assessment, Research & Evaluation

TYPE OF INTRODUCTIONS



The problem statement states a specific condition (issue) that needs urgent attention and a possible solution. As a researcher, show your readers how your investigation will fill or narrow the gap in existing literature. An excellent problem statement is just a line or two. The rest of the paragraph(s) is its elaboration, discussing a possible solution and most importantly, why it is a problem that needs urgent attention (cite scholarly references). The problem must generate questions for the researcher to answer (Faryadi, 2012).

A PhD thesis problem statement must seek to answer the following questions:

- 1) **What** is the problem?
- 2) **Where** is the problem?
- 3) **How** can the problem be solved?
- 4) **Why** do you want to solve the problem?
- 5) **Is** the problem a current issue?
- 6) **Will** the problem persist if it is not solved?
- 7) **Who** are affected adversely by the problem?
- 8) **Will** this problem prove or disprove existing knowledge?

In short, the problem statement describes an existing issue which is so grave that it must be addressed. Generate questions about your problem statement and try to answer those questions to prove or disprove your research problem. Hence, the research problem is the main part of any scientific enquiry (Jacobs, 2013).

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Source: Faryadi, Q. (2018). PhD Thesis Writing Process: A Systematic Approach—How to Write Your Introduction. *Creative Education*, 9, 2534-2545. <https://doi.org/10.4236/ce.2018.915192>

TIPS FOR WRITING THE INTRODUCTION



- 1) State the problem or phenomenon to be investigated.
 - 2) Identify the party affected by the problem.
 - 3) Explain how you plan to solve the problem.
 - 4) Convince the reader that you are qualified and equipped with the right methods of solving that problem.
 - 5) Highlight the benefits of solving the problem.
 - 6) Tell the reader what results you anticipate.
- a) As evident from the above, in your introduction, you should communicate the rationale of your research. Explain the importance of your research. Even though your research introduction chapter has no word limit, being concise helps your readers to comprehend quickly the major issues in your research. When you eventually start writing the introduction, start with some relevant general statements before gradually narrowing it down to focus on the crucial issues such as your research problem, questions, objectives and hypothesis. Here, you explain briefly your research problem and how you plan to solve it. Your introduction must be motivating and captivating enough so that the reader will want to read on to find out more.
- b) It is assumed that your introduction is prepared for readers who have adequate knowledge of your discipline. Your introduction includes many crucial aspects of your thesis. Readers may want to know whether it is your own discovery, or you are continuing a previous study. If it is your own scientific discovery, then you must make it clear how the findings will add to the existing knowledge (Kafes, 2018). Your reader would also want to know more about the concepts used in your research, the objectives of the research, and the methodology employed. You could also discuss the obstacles encountered, if any, and the limitation of your research.



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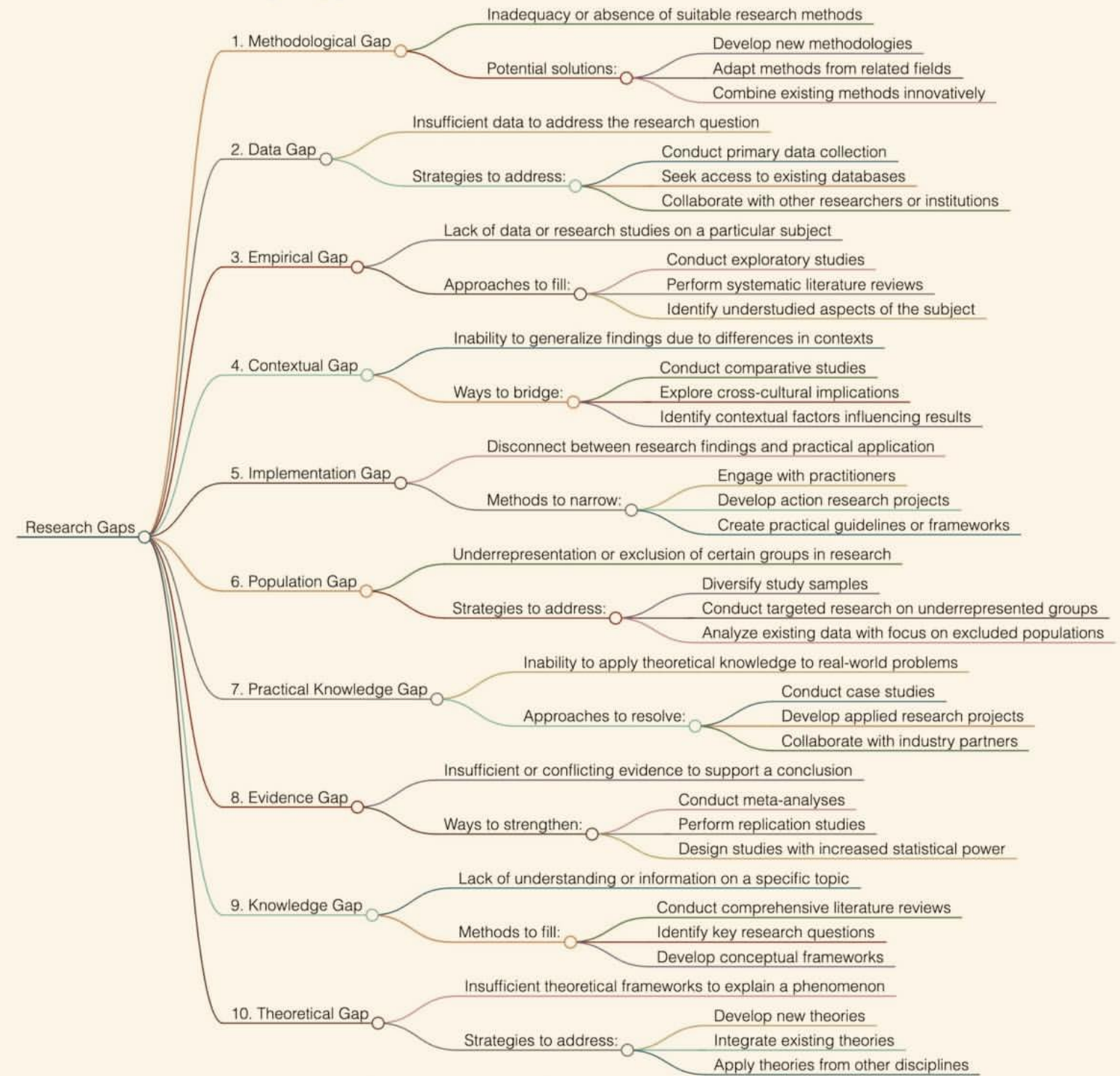
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10 Research Gap Types



Expanded by Lennart Nacke from Miles, D. A. (2017, August). A taxonomy of research gaps: Identifying and defining the seven research gaps. In Doctoral student workshop: finding research gaps-research methods and strategies, Dallas, Texas (pp. 1-15). Mindmap by Lennart Nacke, @acagamic.

CRAFTING A STRONG ABSTRACT

(An Annotated Example from *Nature*)

general background

specific background

knowledge gap

here we show...

results with key, concrete values

meaning of results

Horizontal gene transfer (HGT), the acquisition of genetic material from non-parental lineages, is known to be important in bacterial evolution^{1,2}. In particular, HGT provides rapid access to genetic innovations, allowing traits such as virulence³, antibiotic resistance⁴ and xenobiotic metabolism⁵ to spread through the human microbiome. Recent anecdotal studies providing snapshots of active gene flow on the human body have highlighted (the need to determine) the frequency of such recent transfers and the forces that govern these events^{1,5}. Here we report the discovery and characterization of a vast, human-associated network of gene exchange, large enough to directly compare the principal forces shaping HGT. We show that this network of 10,770 unique, recently transferred (more than 99% nucleotide identity) genes found in 2,235 full bacterial genomes, is shaped principally by ecology rather than geography or phylogeny, with most gene exchange occurring between isolates from ecologically similar, but geographically separated, environments. For example, we observe 25-fold more HGT between human-associated bacteria than among ecologically diverse non-human isolates ($P = 3.0 \times 10^{-270}$). We show that within the human microbiome this ecological architecture continues across multiple spatial scales, functional classes and ecological niches with transfer further enriched among bacteria that inhabit the same body site, have the same oxygen tolerance or have the same ability to cause disease. This structure offers a window into the molecular traits that define ecological niches, insight that we use to uncover sources of antibiotic resistance and identify genes associated with the pathology of meningitis and other diseases.

The human body is a complex biological network comprising ten

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Source: Annotated abstract of a microbiology paper published in Nature.

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